Hongru Zhao Updated October, 2026

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Academic Appointments

IRSA Faragher Distinguished Postdoctoral Fellow, School of Statistics, University of Minnesota 2025–2027

Education

Ph.D. in Statistics (2025), University of Minnesota,
Adviser: Adam J. Rothman
M.S. in Mathematical Sciences (2020), University of Minnesota,
B.S. in Statistics (2018), Jilin University, China

Research interests

Quantum Information Science (theory, advantage, and tomography); High-Dimensional Statistics; Statistical Learning and Generative AI; Astrophysics and Cosmology; Non-Convex Optimization.

Research

— Selected manuscripts and preprints
* Undergraduate research mentee under my supervision.

22. Hongru Zhao and Robert Oliver VandenBerg. (2026) Adaptive quantum tomography by least squares with optimal asymptotic covariance. Preprint. Lean verified.
21. Hongru Zhao. (2026) Minimax Quantum State Tomography with Periodic Clifford Measurements. Preprint. Lean verified. GitHub.
20. Hongru Zhao. (2026) Local Anticoncentration for Gaussian Boson Sampling via Conditional Wishart Geometry. Preprint. Lean verified. GitHub.
18. Hongru Zhao. (2026) Uniform Hiding and Two Routes to Relative Accuracy in Gaussian Boson Sampling. Preprint. Lean verified. GitHub.
14. Hongru Zhao. (2026) On the Log Determinant of Sample Correlation Matrices under Gaussianity. Preprint.
12. Hongru Zhao and Huiqian Feng*. (2026) Convex Reparameterization and Self-Concordant Algorithms for Multivariate Regression with Covariance Estimation. Preprint. GitHub.
11. Hongru Zhao, Jinwen Fu, Tuan Pham. (2025) Convergence and Stability Analysis of Self-Consuming Generative Models with Heterogeneous Human Curation. Preprint.
10. Hongru Zhao. (2025) Allocation-Constrained Adaptive Design and Sequential Estimation: Theory and Application to Bradley–Terry–Davidson Models. Preprint. GitHub.

— Peer-reviewed publications and accepted papers

† Authors are listed alphabetically; author order does not indicate relative contribution.

9. Hongru Zhao, Aaron J. Molstad, and Adam J. Rothman. (2026) Subspace Decompositions for Association Structure Learning in Multivariate Categorical Data Analysis. Electronic Journal of Statistics, accepted. GitHub.
8. Hongru Zhao and Xiaotong Shen. (2026) Distributed Algorithms for High-Dimensional Statistical Inference and Structure Learning with Heterogeneous Data. Statistica Sinica, forthcoming in 38(1) (January 2028).
7. Xiangyu Zhang, Erik Floden, Hongru Zhao, Sara Algeri, Galin Jones, Vuk Mandic, and Jesse Miller. (2026) Validating Angular Power Spectral Models for the Stochastic Gravitational-Wave Background Without Distributional Assumptions. Physical Review D, 113(2), 023051.
6. Sara Algeri, Xiangyu Zhang, Erik Floden, Hongru Zhao, Galin L. Jones, Vuk Mandic, and Jesse Miller. (2026) Testing models for angular power spectra: A distribution-free approach. Physical Review D, 113(2), L021306.
5. Zhibo Zhang, Bixiong Li, Zhiwen Wang, Lianghui Li, Qingshun Nong, and Hongru Zhao. (2026) Microstructure-sensitive modeling of foam concrete based on pore-scale effect quantification and nonlinear assessment of its multifunctional performance. Construction and Building Materials, 537, 147017.
4. Xiaoou Li and Hongru Zhao†. (2025) Globally-Optimal Greedy Active Sequential Estimation. IEEE Transactions on Information Theory, 71(5), 3871–3924. GitHub.
3. Hongru Zhao, and Jinchao Xu. (2024) Convergence Analysis and Trajectory Comparison of Gradient Descent for Overparameterized Deep Linear Networks. Transactions on Machine Learning Research, ISSN: 2835-8856. GitHub.
2. Kangsheng Liu, Zhuangyi Liu, and Hongru Zhao†. (2024) Exponential stability of the linear KdV-BBM equation. Discrete and Continuous Dynamical Systems-B, 29(3), 1206-1216.
1. Yongcheng Qi, and Hongru Zhao†. (2021) Limiting empirical spectral distribution for products of rectangular matrices. Journal of Mathematical Analysis and Applications, 502(2),125237.

— Additional manuscripts and preprints

19. Hongru Zhao. (2026) Shifted Anticoncentration for Real Gram Hafnians and Symmetric Gaussian Hafnians. Preprint. Lean verified. GitHub.
17. Hongru Zhao. (2026) Weak Typicality of von Neumann Entanglement Entropy in Gaussian Boson Sampling. Preprint. Lean verified.
16. Hongru Zhao. (2026) Exact Moments of Gaussian Gram Hafnians Reveal an $n^2/\log n$ Threshold for Weak Anticoncentration. Preprint. Lean verified.
15. Hongru Zhao. (2026) Sharp Berry–Esseen Bounds for the Log Determinant of a Gaussian Sample Correlation Matrix. Preprint. Lean verified. GitHub.
13. Hongru Zhao and Galin Jones. (2026) A Counterexample to Karlin's Conjectured Converse for Admissibility in Exponential Families. Preprint.

Formal Verification Projects

Lean 4 verification of the KLS conjecture with Poincaré constant at most 15 (best known to my knowledge, October 2026). KLS15.

Mentoring & Supervision

— Undergraduate Research

Huiqian Feng* — Undergraduate Research Mentee (high-dimensional statistics), School of Statistics, UMN Fall 2025 – Spring 2026

Invited presentations

Invited Talk and Scholarly Exchange, School of Physics and Astronomy, Sun Yat-sen University, China: Statistical Inference for the Statistical Isotropy Stochastic Gravitational Wave Background (2026).

Invited Talk, Center for Applied Mathematics, Tianjin University, China: From Fisher Information to Adaptive Experimental Design: Statistical Optimality in Quantum State Tomography (2026).

Contributed Paper Presentation, Joint Statistical Meetings (JSM), Nashville, TN: Subspace Decompositions for Association Structure Learning in Multivariate Categorical Response Regression (2025).

Poster Presentation, IRSA Conference, School of Statistics, University of Minnesota: Structure Learning in Multivariate Categorical Response Regression (2024).

Colloquium Seminar, Computer, Electrical, and Mathematical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia: Exploring the Optimization Landscape and Training Dynamics of Deep Linear Networks (2023).

Professional service

— Departmental Service

Seminar Coordinator, UMN School of Statistics Seminar Series, Fall 2025

— External Outreach and Sponsorship

Sponsor representative, WiADS Conference — Represented the IRSA; staffed the sponsor desk, and promoted IRSA consulting and statistical workshops.

— Reviewer Service

Journal of Machine Learning Research (2026); Journal of the American Statistical Association (2026); IEEE Transactions on Information Theory (2025); Electronic Journal of Statistics (2025); Annals of Applied Probability (2022); Transactions on Machine Learning Research (2024–2025).

Courses taught

— UMN-TC: Instructor of Record

Fall 2026: STAT 4052 Statistical Machine Learning II

Spring 2026: STAT 4052 Statistical Machine Learning II

Fall 2025: STAT 4102 Theory of Statistics II

Spring 2025: STAT 3301 Regression and Statistical Computing

Fall 2024: STAT 4101 Theory of Statistics I

Spring 2024: STAT 3301 Regression and Statistical Computing

Fall 2023: STAT 3011 Introduction to Statistical Analysis

— UMN-TC: Graduate Teaching Assistant

Fall 2022: STAT 5101 Theory of Statistics I

Spring 2022: STAT 5102 Theory of Statistics II

Fall 2021: STAT 3021 Introduction to Probability and Statistics

Programming Skills

Lean, Mathematica, MATLAB, Python, R.